Uncovering biological relationships through morphological profiling: The JUMP-Cell Painting dataset and tools

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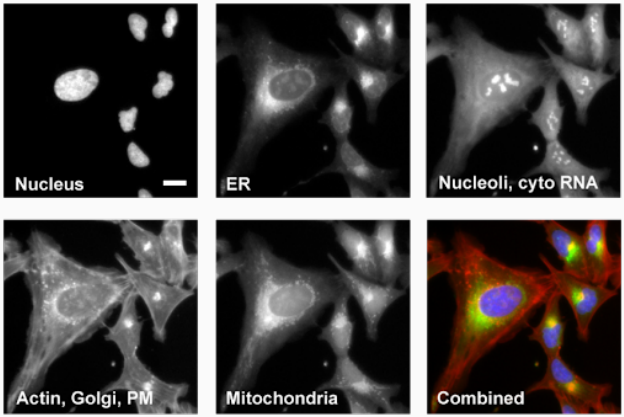
10 November 2026

What is Cell Painting?

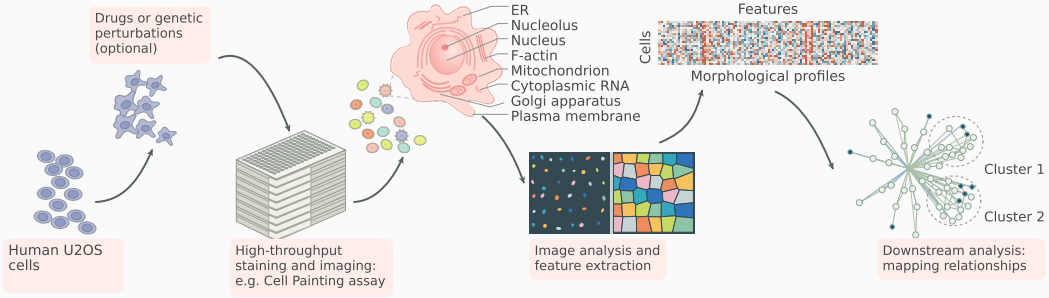
- Fluorescence microscopy of cell morphology.
- Six dyes, five imaging channels.
- Shape, intensity and texture reveal cellular responses.



Source: Muñoz, I2K (2024); assay: Bray et al. (2016)

Cell Painting turns images into measurements

Profiles are numerical descriptions of cells.



Adapted from Chandrasekaran et al. (2021), Fig. 1a

JUMP: Joint Undertaking for Morphological Profiling

- Public Cell Painting data: 12 laboratories, around 136,000 perturbations.
- 1.8 billion cells; ~58.3 million images.

CRISPR knockout	Gene overexpression (ORF)	Compound treatment
7,975 gene targets	**12,609** genes	**115,794** compounds

We make JUMP accessible via the JUMP Hub

Choose Your Own Path

1. Just exploring the data?
   - Use a web interface; no programming required. You can look up images for a gene or compound, distinguishing features, most-similar genes or compounds, and more! See the Interactive Tools page for instructions for these tools.
2. Want to fetch and analyze data?
   - Follow our how-to guides for common analyses
3. Looking to build tools?
   - Use our Python libraries cp_measure, cpgdata, jump-portrait, jump-babel and other libraries in the monorepo
   - Access the metadata schema

Biologists can find out:

- Does my gene or compound have a phenotype?
- Which perturbations cause similar phenotypes?
- What changes in cells and organelles?
- Is the phenotype heterogeneous across cells/samples?

Source: JUMP Hub (2026)

Tutorials, how-tos and resources

The Hub contains practical guides to help biologists use JUMP.

How-to guides	No-code gene/compound queries; explore similarity heatmaps.
Tutorials	Curated runnable notebooks for profiles, metadata and statistical analysis.
Further resources	Dataset explanations, protein-localisation analysis and a Guide to Cell Painting for Biologists.

A tale of two genes (that shape mitochondria)

OPA1: mitochondrial fusion

- Helps mitochondria fuse.
- Loss is expected to produce shorter, fragmented networks.

DNM1L / DRP1: mitochondrial division

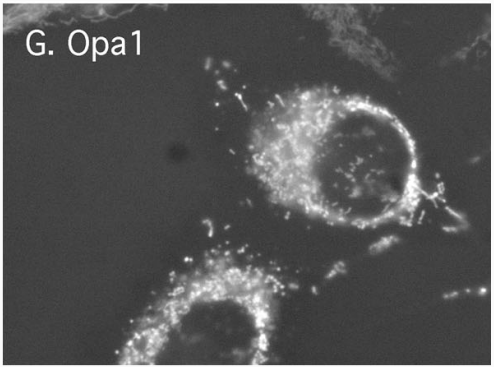
- Helps mitochondria divide.
- Loss is expected to produce longer, more-connected networks.

We will compare knockouts and their negative controls.

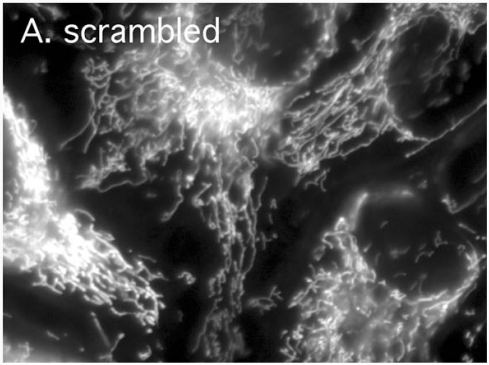
From Griparic et al., JCB (2007), Fig. 3; Smirnova et al., MBoC (2001).

OPA1 depletion fragments mitochondria

OPA1 siRNA



Scrambled siRNA

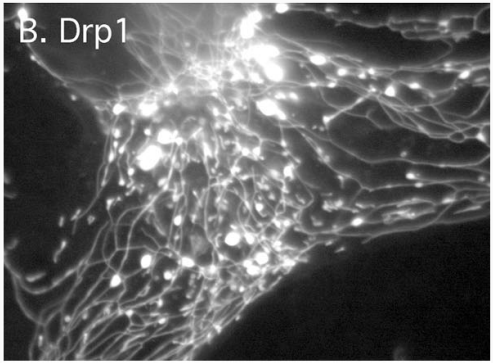


HeLa cells; MitoTracker. OPA1 depletion for 2 days.

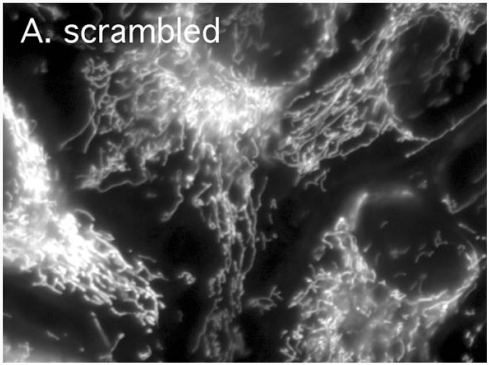
Griparic, Kanazawa & van der Bliek, JCB (2007), Fig. 3G/A. cropped

DRP1 depletion increases connectivity

DNM1L / DRP1 siRNA



Scrambled siRNA



HeLa cells; MitoTracker. DRP1 depletion for 2 days.

Griparic, Kanazawa & van der Bliek, JCB (2007), Fig. 3B/A. Published microscopy, cropped.

What do we want to find in JUMP?

Find the perturbations for OPA1 and DNM1L.

- Start with the gallery and find matching negative controls.
- Follow the gene and reagent into the feature tables.
- Find related perturbations through profile similarity.

How: explore images, investigate features, then follow the matches.

JUMP Hub gene availability

Is the gene represented in JUMP?

10 entries per page Search: OPA

Gene	CRISPR	ORF	A549	HeLa_DMEM	H
str	i64	i64	i64	i64	
COPA	0	1	0	0	
OPA1	1	1	1	0	
OPA3	1	1	0	0	
OPALIN	0	1	0	0	

Showing 1 to 4 of 4 entries (filtered from 15,696 total entries) « ‹ 1 › »

How: search OPA, identify OPA1 among the returned genes, and check which CRISPR and ORF perturbations we can follow into the feature tables.

How do the cells look?

CRISPR Gallery — OPA1

Link	rowid ▼	Perturbation	Plate	Well	Site 0	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8
10049	10049	OPA1	CP-CC9-R1-06	D20									
18309	18309	OPA1	CP-CC9-R2-06	D20									
26592	26592	OPA1	CP-CC9-R3-06	D20									
34225	34225	OPA1	CP-CC9-R4-06	D20									
42436	42436	OPA1	CP-CC9-R5-06	D20									

Tip: enlarge image previews on hover with Imagus (Firefox) or Hover Zoom+ (Chrome).

Filtering example: one OPA1 well

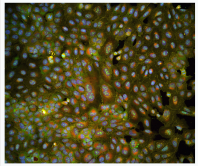
1 row where Perturbation = "OPA1" and Plate = "CP-CC9-R1-06" sorted by rowid

Perturbation	=	OPA1
Plate	=	CP-CC9-R1-06
- column -	=	

Apply

✎ View and edit SQL

This data as json, CSV (advanced)

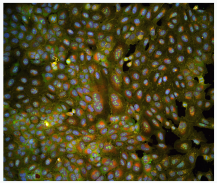
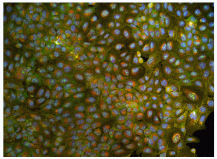
Link	rowid ▼	Perturbation	JCP2022	Site 0	Source	Plate	Well
10049	10049	OPA1	JCP2022_804796		source_13	CP-CC9-R1-06	D20

How: filter the image table for OPA1, then record source_13, plate CP-CC9-R1-06 and treatment well D20 before selecting controls.

What should we compare those cells against?

This data as json, CSV (advanced)

Suggested facets: Perturbation, Resources, JCP2022

Link	rowid	Perturbation ▼	JCP2022	Site 0	Source	Plate	Well
10049	10049	OPA1	JCP2022_804796		source_13	CP-CC9-R1-06	D20
155	155	no-guide	JCP2022_800001		source_13	CP-CC9-R1-06	K02

How: retain source_13 and CP-CC9-R1-06, then add no-guide and non-targeting controls from that same plate.

Which table columns answer our question?

Compartment
Mask used to calculate the feature. It can be Nuclei, Cytoplasm or Cells (the union of both Nuclei and Cytoplasm).

Feature
Morphological feature obtained from CellProfiler. This value is the result after data normalization. Its units are the number of median absolute deviations (MAD) from the median.

Channel
Image channel, which shows the stain for DNA, Mito (mitochondria), RNA, AGP (actin, Golgi, plasma membrane) or ER (Endoplasmic Reticulum).

Suffix
Suffix associated with a CellProfiler feature.

Feature significance
Statistical significance of the difference between a morphological feature in a perturbation compared to the control condition. Lower values suggest a stronger effect of the perturbation on that particular feature. The p-value goes through the Benjamini-Hochberg FDR correction

Cohen's d
Standardized effect size (mean difference / pooled SD). Positive = treatment > control. Sample-size independent, unlike the t-statistic.

|Cohen's d|
Absolute value of Cohen's d. Sort descending to find the strongest effects regardless of direction.

How: read the OPA1 row's reagent, compartment, feature, channel, per-feature significance and signed effect size.

How do we isolate the gene we care about?

31 rows where Perturbation = "OPA1" sorted by Feature Rank

Perturbation = OPA1

- column - =

Apply

✎ View and edit SQL

This data as json, CSV (advanced)

Link	rowid	Compartment	Feature	Channel	Suffix	Feature significance	Cohen's d	|Cohen's d|	I
243374	243374	Cytoplasm	Texture_Correlation	Mito	_3_00_256	0.0	3.56	3.56	

How: filter by the exact symbol OPA1 or a published synonym such as MGM1, then add object-level filters when needed.

Does the perturbation have a measurable phenotype?

31 rows where Perturbation = "OPA1" sorted by Feature Rank

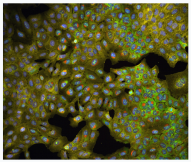
Perturbation = OPA1

- column - =

Apply

✎ View and edit SQL

This data as json, CSV (advanced)

	Perturbation	Corrected p-value	Phenotypic activity	Perturbation example image	Median	Perturbation Rank	Feature Rank ▼	JCP2022
	OPA1	0.00018	0.85692		2.689	99999	0	JCP2022_80

How: read the perturbation-level values: OPA1 CRISPR $p = 0.00018$, activity 0.85692; OPA1 ORF $p = 0.03371$, activity 0.70875.

What does one feature result describe?

OPA1 CRISPR: one feature, read across two close-ups

Compartment	Feature	Channel
Cytoplasm	Texture_Correlation	Mito

Suffix	Feature significance	Cohen's d
_3_00_256	0.0	3.56

How: read across one OPA1 row: reagent, object, feature, channel, significance, effect size and rank. It summarizes a profile, not one cell.

Which features explain the differences?

OPA1 CRISPR: the top-ranked feature

Compartment	Feature	Channel
Cytoplasm	Texture_Correlation	Mito

Suffix	Feature significance	Cohen's d
_3_00_256	0.0	3.56

How: sort OPA1 CRISPR rows by feature rank and compare significance with signed effect size; the leading row is a Mito texture feature.

CellProfiler detects mitochondrial feature changes

Gene	Feature	Cohen's d
OPA1	Mito texture correlation	$+3.56$
DNM1L	Mito-skeleton branch ends	-5.95
DNM1L	Mito granularity	$+4.09$
DNM1L	Mito-skeleton total length	-3.55

Mitochondrial organization changes.

What happens with a different perturbation type?

OPA1 ORF: a separate perturbation and feature ranking

Compartment	Feature	Channel
Nuclei	Granularity_4	Mito

Feature significance	Cohen's d
0.03527	1.406

How: open OPA1 in the ORF table and compare its separate reagent, activity and ranked feature effects with CRISPR.

Which other perturbations have similar profiles?

50 rows where Perturbation = "OPA1" sorted by Perturbation-Match Similarity descending

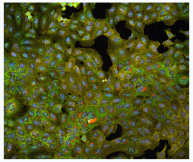
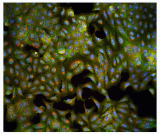
Perturbation = OPA1

- column - =

Apply

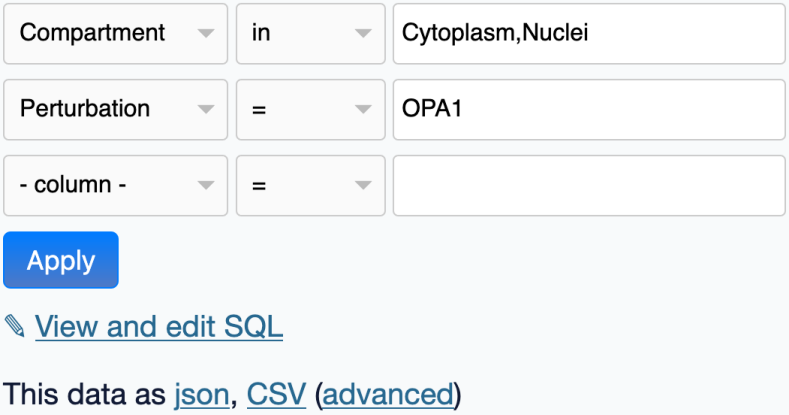
✎ View and edit SQL

This data as json, CSV (advanced)

Link	rowid	Perturbation	Match	Perturbation example image	Match example image	Perturbation-Match Similarity ▲	Match resources
116095	116095	OPA1	MFN1			0.424	NCBI, OMIM, GeneCards, Ensembl

How: filter the CRISPR match table for OPA1 and sort similarity descending; MFN1 is the strongest published match shown (0.424).

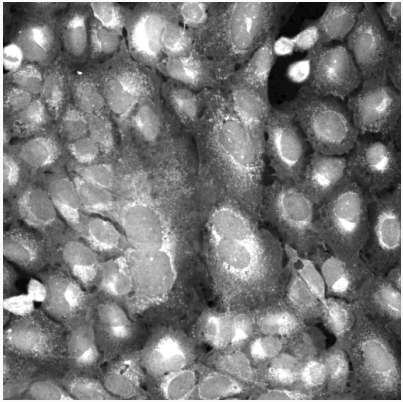
How do we keep and share the result?



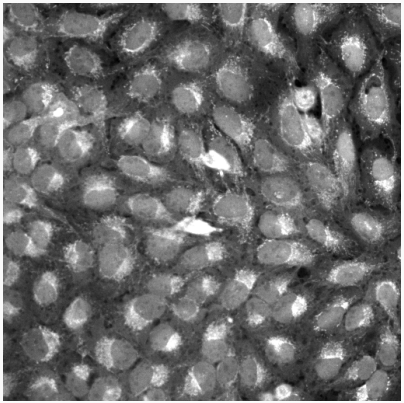
How: keep OPA1 and the relevant objects, then export the rows with the reagent ID, filters and source URL.

JUMP example: OPA1 knockout

OPA1 CRISPR knockout



No-guide negative control

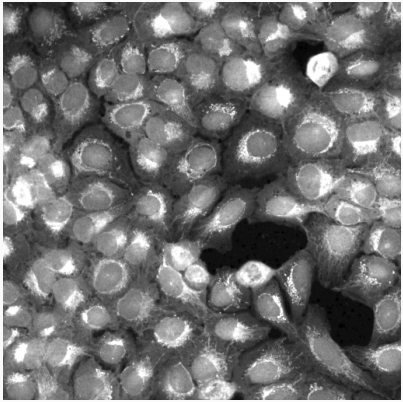


JUMP Mito channel; source_13, CP-CC9-R1-06, D20 / B02, field index 0.

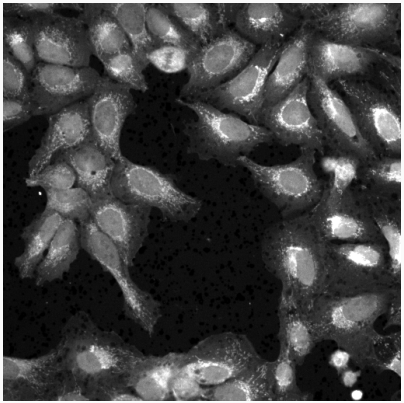
Both panels share intensity limits 219–3985. JUMP data.

JUMP example: DNM1L knockout

DNM1L CRISPR knockout



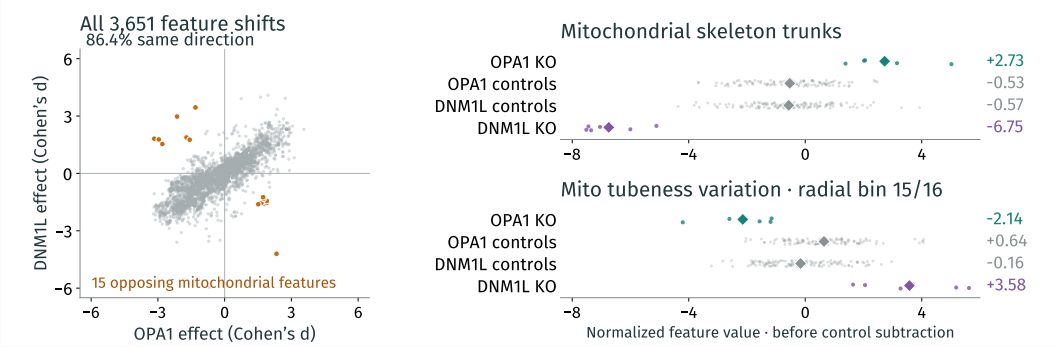
No-guide negative control



JUMP Mito channel; source_13, CP-CC9-R1-02, P18 / B02, field index 0.

Both panels share intensity limits 187–4463. JUMP data.

we can validate existing knowledge to develop methods leading to new discoveries



Shared overall shifts; opposite mitochondrial changes around the controls.

Dots: individual wells. Diamonds: group means. Each KO uses its own plate-matched negative controls.

Agents can help to get to the right data quickly

Start with a biological question

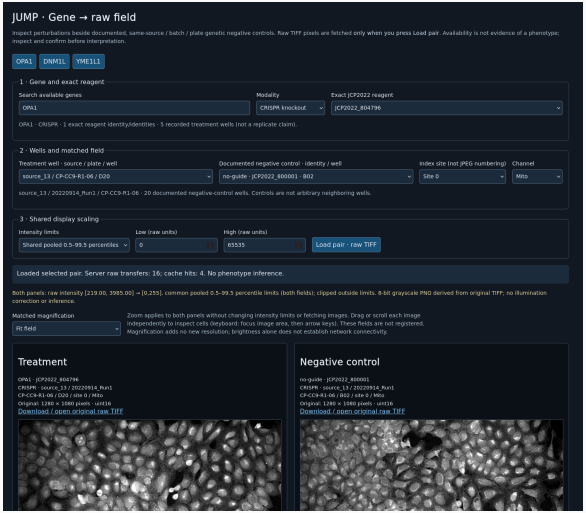
How does disrupting mitochondrial fusion or division change cell images? Find primary studies for OPA1 and DNM1L and explain the expected phenotypes.

Build a tool to inspect the data:

“Let me choose a gene KO, load its raw images beside a negative control in its plate, and zoom into cells using shared intensity limits.”

Output: Interactive viewer.

AI-assisted tools: gene and raw-image explorer

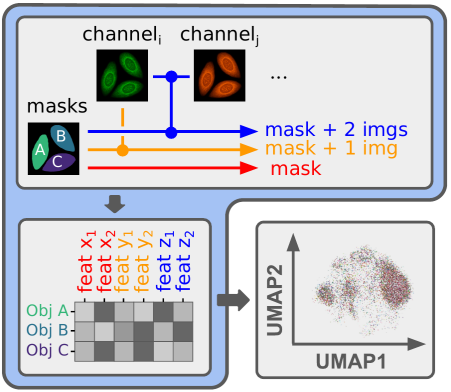


Explorer images: JUMP.

From a gene to its images

- Select a gene and perturbation.
- Load raw images with plate-matched controls.
- Zoom into cells with shared intensity limits.
- Record which differences are visible.

Not just JUMP: Extract morphological profiles from your own images



cp_measure: feature extraction in Python

- Thousands of features across channels and compartments.
- Fits your Python pipeline.
- Fast software to obtain interpretable morphological features (∼5x CellProfiler).

Adapted from Muñoz et al. (2025), Fig. 1

Modern tools make image analysis more accessible

Find cells, nuclei and spots	Cellpose (cells and nuclei); Spotiflow (spots)
Follow cells and organelles over time	Trackastra / Ultrack
Measure cell and organelle appearance	cp_measure
Explore your results	marimo / custom app

Applications of JUMP and our tools across biological questions

Evaluate new lysosomal fluorescence probes

Kenneth Li, University of Edinburgh

- Lysosomal acidification probes in cardiomyocytes.
- Uptake, localization and morphology over time.

Compare imaging modalities

Minh Doan, GSK Bioimaging Analytics

- Bright-field time series, Cell Painting, cell cycle and viability.
- Which dyes and imaging methods best distinguish different mechanisms of action?

MisLocus: protein mislocalisation

Johan Fredin, Carpenter-Shen Lab, Purdue University

- Which rare genetic mutations change protein location?
- Compare microscopy of reference and variant proteins.

Integrated in spatial omics software

Tim Treis, Theis Lab, Helmholtz-Munich

- Integration of morphological features with Squidpy.

Take-home messages

- Known biology helps us identify informative features and develop methods for discovery.
- JUMP provides public Cell Painting data for around 136,000 chemical and genetic perturbations.
- Biologists can use JUMP Hub to find tested genes and compounds, compare phenotypes and inspect images.
- Open-source image-analysis tools extend morphological profiling to your own experiments.
- Guides and notebooks help scientists or AI agents build analysis workflows.

Thank you

Carpenter-Singh Lab
Anne E. Carpenter
Shantanu Singh

Cimini Lab
Beth Cimini
Paula Llanos

Carpenter-Shen Lab
Anne E. Carpenter
Runxi Shen

JUMP partners

JUMP supporting partners

JUMP collaborators

Scientific:
Horizon Discovery,
a PerkinElmer Company
Administrative:
Pistoia Alliance
Academic:
Umeå University
Stanford Machine Learning Group

Work on cp_measure and JUMP-lite was supported by GSK and NIH MIRA (R35 GM122547).

JUMP Hub and slides

JUMP Hub

broad.io/jump

Slides (PDF)

References

All named tools and models below are clickable.

Assay, data and resources

- Cell Painting (Bray, 2016); profiling review (2021).
- JUMP consortium; counts; centers; images.
- JUMP-lite paper; data/code.
- JUMP Hub; how-to guides; notebooks.
- Guide to Cell Painting.

Image-analysis toolbox

- cp_measure; paper.
- CellProfiler; paper.
- Cellpose; StarDist.
- Spotiflow: spot detection.
- Trackastra; Ultrack.
- marimo; uv.
- Squidpy; Morpheus.

Morphological profiling: How to turn images into profiles

Start with a simple question: are control and perturbed cells different?

Classical features (e.g., CellProfiler)

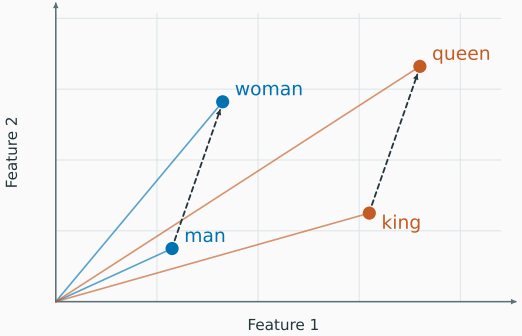
- Size, shape, intensity, texture and relationships between channels and objects.
- Interpretable but partially redundant.

Learned representations

- Features learned from images by neural networks (deep learning).
- Non-redundant numerical vectors (embeddings).

Words as vectors: word2vec

Learning can be viewed as compression



Word → vector

Learned from context

man → woman
king → queen

$$v(\text{king}) - v(\text{man}) + v(\text{woman}) \approx v(\text{queen})$$

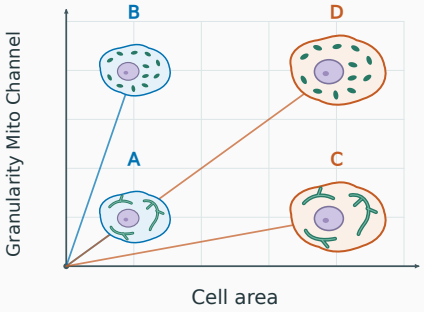
Mikolov et al. (2013)

From words to cells: vectors for comparison

Classical features

Defined measurements

x = [cell area, granularity, ...]

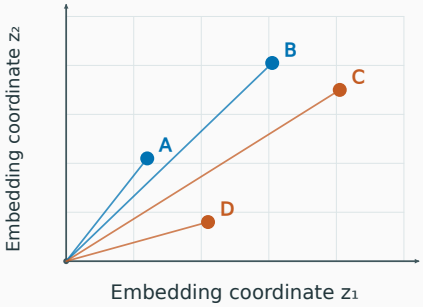


Learned embeddings

Learned from images

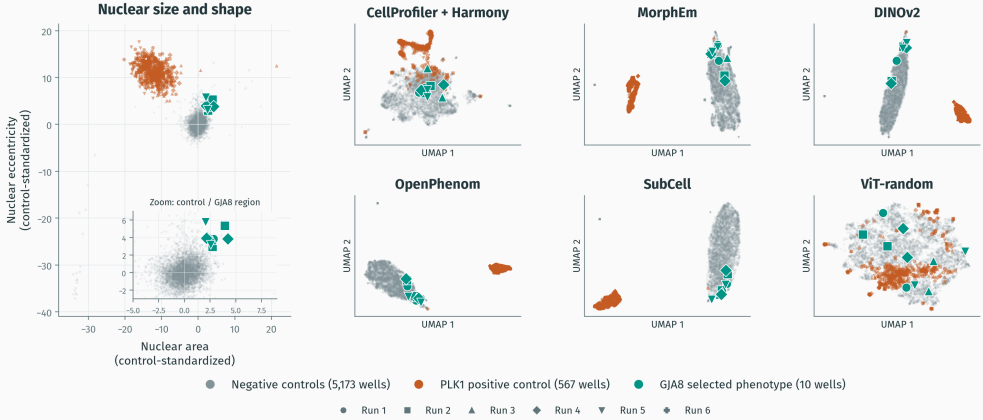
Not necessarily directly interpretable.

z = [z_1, z_2, ..., z_n]



● Control ● Perturbation

Positive controls contextualize perturbations in both learned and classical representations



Data: JUMP-lite (2026); models: MorphEm, DINOv2, OpenPhenom, SubCell; CellProfiler/Harmony recipe

JUMP-lite: a compact benchmark

Same images. Different methods.

Test whether profiles recover known relationships.

Same selected images: 8.7 TB $\rightarrow$ 92 GB ($\sim$94$\times$ smaller; lossy JPEG XL).

Adapted from Muñoz et al. (2026)

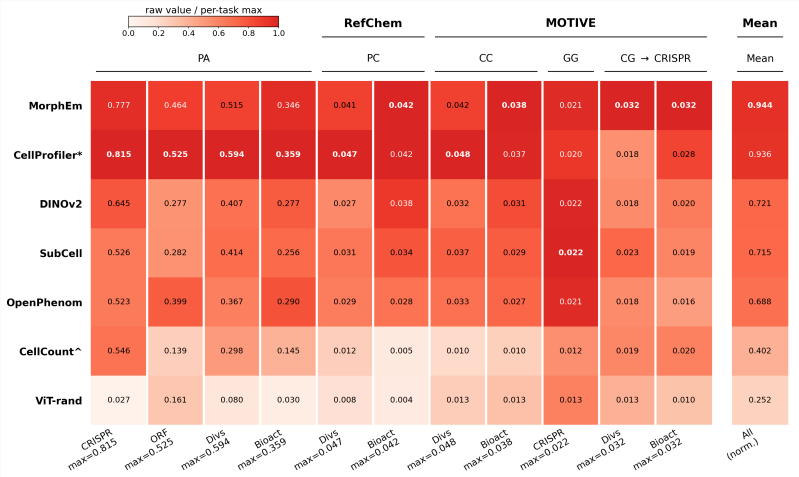
Nahual: swap models within one workflow

Swap models. Keep your workflow.

Image features	DINOv2 ↔ DINOv3 ↔ OpenPhenom ↔ MorphEm
Segmentation	Cellpose ↔ StarDist
Spot detection	Spotiflow
Tracking	Trackastra ↔ Ultrack

JUMP-lite: comparing image representations

Recover replicate perturbations and known biological relationships.

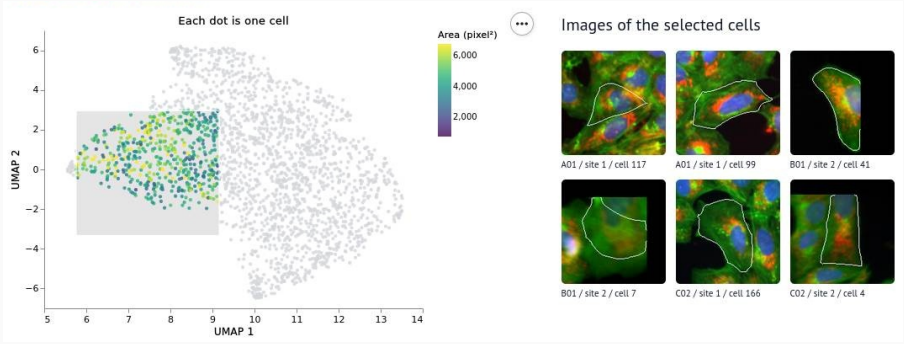


Raw images; colors normalized within each task. CellProfiler/count references use more sites per well.

Adapted from Muñoz et al. (2026), Figure 5

Backup: explore cell subpopulations

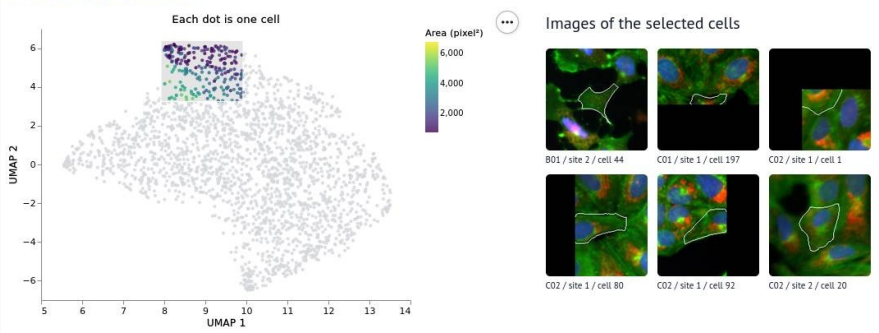
Interactive marimo notebook — separate recorded example



Each dot is a cell: 2,223 from one Target-2 plate. Shown selection: 545 cells. Inspect images, area and other features.

Source: Muñoz (2026), JUMP Target-2 notebook recording

Backup: compare another cell selection



- Second selection: 224 cells; compare images and measurements.
- Not the compound queries; no treatment/control assignments in this example.

Source: Muñoz (2026), JUMP Target-2 notebook recording